

Models, Tools, and Methods Report — Darian Ding

Project: Katmai CV Pipeline — Bear Detection, Tracking, and Salmon Feeding Behavior Analysis **Date:** April 26, 2026 **Author:** Darian Ding (Annotation Pipeline, Salmon Detection MoE, Behavior-Analysis VLM Integration)

0. Evaluation Methodology

Two complementary methods were used to compare candidate models throughout the project:

0.1 Public Benchmark Rankings (model shortlisting)

Candidate models were first shortlisted by consulting publicly maintained leaderboards relevant to each task. This let us narrow a large field of open-source releases down to a small set worth bench-testing on Katmai footage.

Task	Benchmarks consulted
Open-vocabulary detection	ODinW (Object Detection in the Wild), COCO zero-shot, LVIS — used to compare Grounding DINO, OWL-ViT, Florence-2, MegaDetector.
Closed-vocabulary detection	COCO mAP@0.5, COCO mAP@0.5:0.95, Roboflow 100 — used to compare YOLOv8 variants, RetinaNet, DETR.
Vision-language models (image)	OpenCompass / OpenVLM Leaderboard, MMBench, MMMU — used to compare Molmo2, LLaVA-OneVision, Qwen2.5-VL, InternVL2.
Vision-language models (video)	Video-MME, MVBench, TempCompass, LongVideoBench — used to compare VideoLLaMA2, LLaVA-OneVision (video), Qwen2.5-VL (video), Molmo2 (video).
Multi-object tracking	MOT17 / MOT20 (HOTA, MOTA, IDF1) — used to compare ByteTrack, BoT-SORT, DeepSORT.

Benchmark rankings were **necessary but not sufficient**: a model that scores well on Video-MME (e.g., on movie clips with clean cuts) does not necessarily perform well on Katmai's static-camera, splash-occluded, animal-only footage. We therefore used the rankings only as a filter to decide which 2–3 models per task were worth running on our own data.

0.2 Manually-Created Ground Truth (final model selection)

For each task we manually constructed a small but representative ground-truth set from Katmai footage and measured candidate models against it. This was the deciding test for every modeling choice in the pipeline.

Ground-truth set	How it was built	What it evaluated
848 frames / 745 bear bboxes	Manually verified subset of the 24,238-frame auto-annotation output. Each frame inspected; false positives removed; missed bears added. Labels exported in YOLO format.	YOLOv8 baseline vs. fine-tuned (Precision / Recall / F1 / mAP@0.5 / mAP@0.5:0.95).
Per-camera-view subsets (Brooks Falls Low, Multiview, Riffles, River Watch)	Stratified sampling from the same labeled set, grouped by source camera.	Whether the fine-tuned detector generalizes across camera angles or overfits to one view.
Bear-count ground truth on full clips (e.g. 2025-09-19 23-30-11_Brooks_Falls_Low_5_bears.mp4 , manually labeled count = 5)	Watched each clip end-to-end and counted distinct bears that appeared.	ByteTrack vs. BoT-SORT — measured by unique_bears_tracked against manual count, and by ID-switch count.
~50-frame manually-labeled bear-feeding sequence	Manual labeling of feeding stage per visible bear per frame: WAITING / LUNGING / CATCHING / EATING / MISSED .	Per-frame VLM accuracy: Molmo2-8B vs. LLaVA-OneVision-7B vs. Qwen2.5-VL-7B vs. InternVL2-8B. Measured stage-match accuracy and bear-ID correctness.
Salmon-jump count on short clips (salmon_jump_0.mp4 ... salmon_jump_9.mov , 5-22 s)	Watched each clip frame-by-frame and recorded the timestamp of each visible jump.	Molmo2 video-mode count vs. ground-truth count; jump-timestamp error.

The combination of (a) benchmark-driven shortlisting and (b) Katmai-specific manual ground truth gave us defensible reasons for every model choice in the pipeline.

1. Models Experimented With

Detection & Auto-Annotation Models

Model	URL	Capability
Grounding DINO	https://github.com/IDEA-Research/GroundingDINO	Open-vocabulary object detector; takes free-text prompts (e.g. "bear", "fish jumping") and returns bounding boxes without task-specific training. SwinT/SwinB backbone fused with BERT text encoder. State of the art on ODinW at the time of selection.
OWL-ViT (v2)	https://huggingface.co/google/owlv2-base-patch16-ensemble	CLIP-based open-vocabulary detector. Strong on small / dense objects via patch-level attention.

Model	URL	Capability
Florence-2	https://huggingface.co/microsoft/Florence-2-large	Microsoft's unified vision foundation model. Multi-task: detection, segmentation, captioning, region description. Performs well on water / reflection scenes.
MegaDetector v6	https://github.com/agentmorris/MegaDetector	Microsoft AI-for-Earth detector trained on millions of camera-trap images. Three classes: animal, person, vehicle.
DETR	https://huggingface.co/facebook/detr-resnet-50	Transformer-based end-to-end detector. Strong at modeling global spatial relationships.
YOLOv8 (n / s / m)	https://github.com/ultralytics/ultralytics	Single-stage real-time detector. Pretrained on COCO (which includes "bear" class 21). Selected as the backbone we fine-tuned.
RetinaNet	https://pytorch.org/vision/stable/models/retinanet.html	Single-stage detector with focal loss for class imbalance; considered as alternative to YOLO.
CLIP ViT-L/14	https://huggingface.co/openai/clip-vit-large-patch14	Image-text contrastive model. Used as the scene-feature extractor for the salmon Mixture-of-Experts gating network (2048-d feature → 3-layer MLP → expert weights).

Vision-Language Models (Behavior Analysis & Salmon Counting)

Model	URL	Capability
Molmo2-8B (selected)	https://huggingface.co/allenai/Molmo2-8B	Allen Institute open-weights VLM. Native video input via chat template. Outputs structured text + point coordinates. 8B parameters; runs on 22 GB VRAM in bf16. Strong on OpenVLM Leaderboard for open-weights models.
LLaVA-OneVision-7B	https://huggingface.co/llava-hf/llava-onevision-qwen2-7b-ov-hf	Single-image, multi-image, and video understanding in one model. Built on Qwen2-7B + SigLIP. Top-tier on Video-MME for 7B class.
Qwen2.5-VL-7B-Instruct	https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct	Alibaba's VLM with native video support, dynamic resolution, and grounding (bbox output). Strong instruction following. Top of MMBench at release.
InternVL2-8B	https://huggingface.co/OpenGVLab/InternVL2-8B	Open-source VLM with strong OCR and multi-image reasoning, video via interleaved frame input.
VideoLLaMA2-7B	https://huggingface.co/DAMO-NLP-SG/VideoLLaMA2-7B	Purpose-built video VLM with spatio-temporal convolutional connector. Strong on MVBench .
GPT-4o	https://platform.openai.com/docs/models/gpt-4o	OpenAI multimodal closed model. Image understanding via API; no native video, must pre-sample frames.
Gemini 1.5 Pro	https://ai.google.dev/gemini-api	Google closed model, 1M-token context, accepts long video natively (~1 hour).

Tracking

Model	URL	Capability
ByteTrack (selected)	https://github.com/ifzhang/ByteTrack	Two-stage motion-only association with Kalman filter. Recovers occluded objects via low-confidence detections. Top of MOT17 / MOT20 at release.

Model	URL	Capability
BoT-SORT	https://github.com/NirAharon/BoT-SORT	Motion + appearance association with camera motion compensation.
DeepSORT	https://github.com/nwojke/deep_sort	Motion + appearance Re-ID. Requires a separate Re-ID network.

2. Models We Created / Modified

Model	What it does
Fine-tuned YOLOv8n bear detector (<code>bear_detector3/best.pt</code>)	YOLOv8-Nano fine-tuned on 24,238 Katmai frames (40,000+ auto-generated bounding boxes from Grounding DINO). Single class: <code>bear</code> . mAP@0.5 = 95.1%, F1 = 91.4% on validation.
Salmon Mixture-of-Experts annotation pipeline	CLIP-ViT-L gating network (3-layer MLP, ReLU + Dropout 0.2 + softmax) routes each frame to a weighted combination of Grounding DINO, OWL-ViT, and Florence-2. Platt scaling calibrates per-expert scores; weighted fusion + IoU NMS produces the final boxes.
Molmo2-8B Bear Feeding Behavior Classifier (prompt-engineered)	Wraps Molmo2 with a domain-specific prompt and ByteTrack ID overlays. Per-frame inference produces a 5-stage classification (<code>[WAITING]</code> / <code>[LUNGING]</code> / <code>[CATCHING]</code> / <code>[EATING]</code> / <code>[MISSED]</code>) per tracked bear, then merged with stage-aware deduplication.
Molmo2-8B Whole-Video Summarizer	Builds a behavior timeline + a downscaled mid-video reference frame, sends both to Molmo2 for a 2–4-sentence narrative summary appended to the demo video.
ByteTrack ID-fragmentation merger	Custom post-processing on top of ByteTrack to merge fragmented IDs of the same bear (<code>_merge_fragmented_tracks</code>) and remap to display IDs 1..N.

3. Pros / Cons of Each Model

Where applicable, the "Pros" column references **public benchmark scores** that drove shortlisting, and the "Cons" column reports what we observed when running each model against our **manually-created Katmai ground truth**.

Detection / Annotation

Model	Pros (benchmark + observed)	Cons (observed against manual ground truth)
Grounding DINO	Best zero-shot ODinW score among open detectors at selection time; ~95%+ accuracy on our 4,848-frame manual bear ground truth in spot-checks; flexible text prompts; handles partial occlusion	Slow inference (~1.5 s/frame on 2080 Ti); memory-hungry; ~3–5% false-positive rate on water-textured backgrounds; mediocre recall on small distant salmon
OWL-ViT	Better recall than Grounding DINO on small dense objects in our salmon spot-check; CLIP-based grounding scales well	More false positives on textured backgrounds (waterfall splash); slower per-frame

Model	Pros (benchmark + observed)	Cons (observed against manual ground truth)
Florence-2	Strong on water / reflection scenes; multi-task (caption + detect); fast	Less precise localization on small objects; prompt format is quirky and required custom parsing
MegaDetector	Trained on wildlife camera-trap data; very fast; works offline; widely cited in conservation ML	Only 3 generic classes (animal/person/vehicle) — cannot distinguish bear from other animals; no salmon class
DETR	Models global context well; strong COCO mAP	Heavy compute; weaker without large task-specific data; mature tooling lacking
YOLOv8 (pretrained, COCO bear)	Real-time; mature ecosystem; trivial fine-tuning	Out-of-the-box recall on Katmai manual ground truth = 1.72% (essentially unusable without fine-tuning)
YOLOv8 (fine-tuned)	Production-ready: precision 92.2%, recall 90.6%, mAP@0.5 95.1% on manual validation set; runs on laptop	Single-class only; no behavior understanding; needs re-training for new species
RetinaNet	Good accuracy / speed balance on COCO	More careful hyperparameter tuning required than YOLO; higher compute cost; ecosystem less mature

VLMs (for feeding behavior + salmon counting)

Each candidate VLM was scored against a **~50-frame manually-labeled bear-feeding ground-truth set** (consensus labels from three team members) and a **set of short salmon-jump clips with manually-counted ground-truth jump counts**.

Model	Pros (benchmark + observed)	Cons (observed against manual ground truth)
Molmo2-8B (selected)	Open weights + commercial license; high score on OpenVLM Leaderboard for open 8B class; strong domain reasoning; native video input via chat template; supports point coordinates and bounding boxes; runs in bf16 on 22 GB VRAM (dual 2080 Ti); follows structured output format reliably; highest stage-match accuracy on our 50-frame feeding set among 7–8B open VLMs we tested	~5 s per frame inference; video mode capped at ~8–10 frames on 22 GB VRAM (frame budget × tokens-per-frame is the bottleneck); occasionally outputs <code><points></code> instead of text without explicit prompt suppression
LLaVA-OneVision-7B	Top-tier on Video-MME for 7B class; strong open-source video VLM; HuggingFace integration; handles multi-image input	On our 50-frame feeding ground truth, behavior labels were ~20% inconsistent vs. manual labels; no native bbox output, so cannot reference bear IDs by drawing
Qwen2.5-VL-7B	Excellent grounding (bbox output natively); dynamic resolution; strong instruction following; top of MMBench at release	Verbose outputs that broke our parser without heavy prompt engineering; slight Chinese-tuned bias in style; 7B model still hit OOM on long videos

Model	Pros (benchmark + observed)	Cons (observed against manual ground truth)
InternVL2-8B	Strong general benchmarks; OCR is excellent	Video support is essentially "interleaved frames" with no native temporal modeling; produced wandering descriptions rather than a clear stage tag; failed our structured-output requirement
VideoLLaMA2-7B	Purpose-built for video; strong on MVBench; spatio-temporal connector	Weaker zero-shot domain knowledge of wildlife in our pilot; required fine-tuning we couldn't afford on lab hardware; less mature ecosystem
GPT-4o	Best raw quality on our 50-frame ground truth; fast API	Closed/paid (per-frame API cost; 24K frames $\times$ N runs is prohibitive); no on-prem inference; data-residency concerns for sponsor-owned video; no native video — must pre-sample frames
Gemini 1.5 Pro	Native long-video input (1M-token context); can ingest entire 15-min clip in one call	Closed/paid API; requires upload of footage to Google; rate-limited; opaque internal sampling strategy gave inconsistent answers across repeated runs on the same clip

Tracking

Compared on a held-out 5-bear test clip (manually-counted ground truth = 5).

Tracker	Pros	Cons
ByteTrack (selected)	No appearance net needed (bears look very similar so Re-ID is unreliable); recovers occluded tracks via second-stage low-conf association; top of MOT17/MOT20; on the 5-bear test clip held 5 stable IDs through splash occlusion (after our ID-merger post-processing)	Reassigns new ID when bear fully exits and re-enters frame; sensitive to confidence-threshold choice
BoT-SORT	Camera-motion compensation helps with zoom transitions	Heavier; appearance branch unreliable on visually-identical bears (we observed 8–10 fragmented IDs on the same 5-bear clip)
DeepSORT	Mature; widely documented	Re-ID extractor needed; appearance features fail for visually identical bears under water splash

4. Criteria Each Model Uses to Analyze Frames / Videos

Model	Decision criteria
Grounding DINO	Cross-modal alignment between BERT-encoded text prompt and Swin-Transformer image features; outputs (bbox, confidence, matched-token) per query
OWL-ViT	CLIP image-text similarity per ViT image patch → patch-level objectness + class score
Florence-2	Sequence-to-sequence: image patches + task prompt → token sequence parsed into structured output (boxes, labels, captions)
YOLOv8	Single-pass anchor-free regression: CSPDarknet backbone → PAN neck → decoupled head producing (x, y, w, h, objectness, class-prob) per grid cell; conf threshold 0.25
CLIP gating network	2048-d ViT-L semantic vector → 3-layer MLP → softmax weights over 3 expert detectors based on scene content (waterfall vs. underwater vs. surface)
ByteTrack	Two-stage Hungarian matching: (1) Kalman-predicted track box vs. high-conf detection by IoU (threshold 0.7); (2) unmatched tracks vs. low-conf detections (threshold 0.15); fuse_score=True blends conf with IoU
Molmo2-8B (per-frame mode)	Chat-template input: image (with ByteTrack bbox + ID overlay) + structured prompt requesting Bear N: [STAGE] description ; greedy decode with do_sample=False
Molmo2-8B (video mode)	Internal video processor uniformly samples up to max_fps × duration frames (default max_fps=2.0, num_frames=384); each frame tokenized and concatenated into a single sequence; cross-attention over all tokens

5. Modifications Made and Outcomes

Grounding DINO

- **Modification:** Wrapped in a custom batch-inference script with confidence filtering, YOLO-format conversion, and a visualization debug module.
- **Outcome:** Generated 40,000+ bounding boxes across 24,238 frames in ~14 hours on dual 2080 Ti — work that would have taken weeks of manual annotation. Spot-check accuracy ~95%; the ~3–5% noise was tolerated by YOLO fine-tuning.

Salmon Mixture-of-Experts

- **Modification:** Built CLIP-gated routing over Grounding DINO + OWL-ViT + Florence-2; added Platt scaling for cross-model score calibration and IoU-NMS for fusion.
- **Outcome:** Higher recall on visually-complex water scenes than any single detector. Ongoing in Spring Quarter; preliminary review queue is enabled for low-confidence frames.

YOLOv8n Fine-Tuning

- **Modification:** Transfer learning from COCO weights; cosine LR schedule (0.01 → 0.0001); 640 × 640 input; mixed-precision; mosaic + horizontal-flip + scale augmentation; rotation/perspective disabled (fixed camera); early stopping with patience=50.

- **Outcome:** Measured against manual ground truth: Precision 34.37% → 92.2% (+168%); Recall 1.72% → 90.6% (+5167%); mAP@0.5 17.87% → 95.1% (+432%). All six camera views converged to >90%.

ByteTrack

- **Modification:** Tuned thresholds for the splash-occlusion scenario: `track_high_thresh=0.4`, `track_low_thresh=0.15`, `new_track_thresh=0.85`, `track_buffer=300-450`, `match_thresh=0.7`, `fuse_score=True`. Added a custom post-processing step `_merge_fragmented_tracks` to collapse fragmented IDs of the same bear and remap to display IDs 1..N.
- **Outcome:** Reduced ID switches dramatically; on the 5-bear manually-counted test clip the tracker holds 5 stable IDs throughout splash occlusion instead of fragmenting into 8–10 IDs.

Molmo2-8B for Bear Feeding Behavior

- **Modifications:**
- **On-frame ID annotation:** Draw ByteTrack bounding box + bear ID on each frame before sending to Molmo2, so the model can reference bears by the same numbering the user sees in the demo video.
- **Stage-tag prompting:** Force structured output `[WAITING] / [LUNGING] / [CATCHING] / [EATING] / [MISSED]` per bear, so dedup logic can detect stage transitions reliably.
- **Stage-aware deduplication:** Replaced naïve `SequenceMatcher` ratio-based dedup with stage-tag extraction — any stage change is always treated as a new event, even if the surrounding text is similar. This fixed missed `CATCHING` events at video end.
- **Whole-video summary pass:** Built a behavior timeline string + a downscaled reference frame (max 512 px) and sent it as a final summary call. `torch.cuda.empty_cache()` before this call avoided OOM at 0.25 s sampling intervals.
- **Outcome:** Bear IDs now match between video and analysis text. Catch events at end-of-clip ("fish in mouth") are no longer swallowed by dedup. Side-by-side viewer renders correctly.

Molmo2-8B for Salmon Jump Counting

- **Modifications:** Tested native video input mode; varied `max_fps`, frame count, and input resolution (1966×1102 → 720p → 480p → 360p) to fit VRAM budget.
- **Outcome: Negative result.** 22 GB of VRAM (dual 2080 Ti) only fits ~8–10 video frames in Molmo2's video mode regardless of resolution — the bottleneck is the per-frame token count multiplied by attention cost. With sparse sampling, the model hallucinates jump events at impossible timestamps (e.g. claiming a jump at 0:15 in a 4-second clip). Counting accuracy is unreliable; we are now exploring a YOLO-based salmon detector + ByteTrack trajectory analysis as a fallback.

6. Best-Performing Models (and Why)

Selection criteria, in priority order: (1) accuracy on **Katmai-specific manual ground truth**; (2) ability to run on consumer hardware (PR-2); (3) open-source / open-weights to avoid recurring API cost and data-residency issues; (4) tooling maturity and HuggingFace / Ultralytics integration for fast iteration.

Task	Best model	Why
Bear detection	Fine-tuned YOLOv8n (bear_detector3/ best.pt)	mAP@0.5 = 95.1%, F1 = 91.4% on the manually-verified validation set; runs at ~30 FPS on a CPU-only laptop and ~200 FPS on the 2080 Ti; satisfies the consumer-grade hardware constraint (PR-2).
Bear auto-annotation	Grounding DINO	Best ODinW benchmark score among open detectors at selection time, and best zero-shot accuracy on our manual bear spot-check set; text-prompt flexibility allowed us to label 24K frames without building a manual labeling team.
Multi-object tracking	ByteTrack (with our fragmentation merger)	Top of MOT17/MOT20 leaderboard; on the 5-bear manually-counted test clip, motion-only association is the right choice when subjects look identical; second-stage low-conf recovery handles splash occlusion.
Bear feeding behavior classification	Molmo2-8B (per-frame, with on-image ID overlay)	Highest stage-match accuracy on our 50-frame manual feeding ground truth among open 7–8B VLMs tested; open weights so we can run on-prem; native bbox/point support; consistent structured output.
Salmon jump counting	Unresolved.	Molmo2 video mode is currently the best demonstrated approach but is hardware-limited.

7. Missing Model Capabilities

Across every model we tried, the following capability gaps materially slowed progress on bear detection, tracking, and salmon-feeding analysis:

1. **Memory-efficient long-video VLM inference.** Every open-weights VLM (Molmo2, LLaVA-OneVision, Qwen2.5-VL, VideoLLaMA2) hits a hard wall around 8–16 video frames on 22 GB of VRAM because per-frame visual tokens $\times O(n^2)$ attention dominate. There is currently no open VLM that supports streaming video understanding (KV-cache eviction, sliding-window attention, or hierarchical token compression) on consumer GPUs. Closed models like Gemini 1.5 Pro have this but at API cost and with data-residency constraints. **If this existed, salmon jump counting and long-form bear behavior analysis would be solvable in a single pass instead of frame-by-frame.**
2. **Cross-instance Re-ID for visually identical animals.** Pretrained Re-ID networks are trained on humans / vehicles and fail on bears that look near-identical. None of the trackers we tested can reliably re-associate a bear that fully exits and re-enters the frame. Solving cross-video bear identification was actually descoped from the project for this reason. A wildlife-specific Re-ID foundation model (analogous to MegaDetector for detection) is missing from the open-source ecosystem.
3. **Native temporal action / event localization.** Detection models output bounding boxes per frame, and VLMs output text per frame, but neither natively outputs (start_time, end_time, event_label, agent_id) tuples. We have to reconstruct events from per-frame outputs with custom dedup and stage-tracking logic. A true video-action-detection foundation model (something like a wildlife-domain ActionFormer or VideoMAE-V2 fine-tuned on animal behavior) would let us treat "salmon catch" as a first-class detected event.
4. **Small-object detection in heavily textured backgrounds.** Salmon in turbulent water (waterfall splash, surface reflections) defeat all three open-vocabulary detectors we tried. Recall is poor and

false positives on splash artifacts are high. A model with explicit motion-cue or optical-flow conditioning would likely help.

5. **Cheap on-prem fine-tuning of large VLMs.** We could not fine-tune Molmo2-8B on Katmai-specific behavior labels because LoRA + 8B + bf16 video training does not fit comfortably on dual 2080 Ti without a heroic engineering effort (gradient checkpointing, ZeRO-3, etc.). Smaller VLMs (~2B) do fit but lose enough reasoning quality to be useless. A 2–3B parameter VLM with Molmo2-level reasoning would change what is achievable on lab hardware.

8. AI Tools Used or Modified

Tool	URL	Pros	Cons / Where it needed supervision
Ultralytics YOLOv8 framework	https://github.com/ultralytics/ultralytics	Single-line train/predict/track API; built-in ByteTrack/BoT-SORT integration; HuggingFace + ONNX export; large community.	Tracker config YAML format was poorly documented — <code>fuse_score: True</code> is required for ultralytics' tracker but not mentioned in the official ByteTrack README, costing hours of debugging. Default <code>vid_stride=1</code> silently processes every frame, catastrophically slow on long video.
HuggingFace transformers (4.57.6)	https://github.com/huggingface/transformers	Unified <code>AutoModelForImageTextToText</code> + <code>AutoProcessor</code> for Molmo2; chat-template handles video loading; <code>device_map="auto"</code> for multi-GPU.	Version churn is a major hazard: transformers 5.5.4 broke Molmo2 with <code>Unexpected keyword argument image_use_col_tokens</code> ; transformers 4.57.6 requires <code>torch>=2.6</code> (<code>Using or_mask_function arguments require torch>=2.6</code>). Pinning was non-trivial.
Grounding DINO (HuggingFace port)	https://huggingface.co/IDEA-Research/grounding-dino-base	Easy text-prompt API; reproducible across machines via HF Hub.	Slow (~1.5 s/frame); occasional CUDA OOM on 1080p frames without explicit downscaling.
CLIP ViT-L (HuggingFace)	https://huggingface.co/openai/clip-vit-large-patch14	Reliable feature extractor; fast on GPU.	Embedding stability across CLIP versions varied; we pinned a specific revision.

Tool	URL	Pros	Cons / Where it needed supervision
OpenCV + ffmpeg	https://opencv.org/ / https://ffmpeg.org	Video decode, frame extraction, trimming, format conversion.	OpenCV <code>VideoCapture</code> on <code>.mov</code> files was unreliable on Linux without ffmpeg backend; XVID writer outputs <code>.avi</code> which we re-encode to <code>.mp4</code> for sharing.
decord	https://github.com/dmlc/decord	Required by Molmo2's video processor; faster than OpenCV for random-access frame loading.	Quietly required — Molmo2 raised <code>ImportError: requires decord, torchcodec, or av</code> and <code>pip install decord</code> fixed it.
CVAT	https://github.com/opencv/cvat	Industry-standard labeling UI; YOLO format export; used to construct manual ground-truth sets.	Heavy to self-host (Docker, multiple services); used sparingly for spot-check verification rather than full annotation due to timeline.
Label Studio	https://github.com/HumanSignal/label-studio	Lighter than CVAT; flexible label schemas; good for the human-in-the-loop review queue in the salmon MoE pipeline and for the 50-frame feeding ground-truth set.	Web UI sometimes lags on long video frames; export format requires post-processing to reach YOLO format.
bitsandbytes	https://github.com/bitsandbytes-foundation/bitsandbytes	4-bit / 8-bit quantization to fit larger VLMs in VRAM.	Requires a working C compiler and recent CUDA; on our ENGINE Lab machine we hit <code>RuntimeError: Failed to find C compiler</code> on Triton compilation and reverted to bf16. Not a drop-in replacement.
PyTorch (2.6.0+cu124)	https://pytorch.org	Required for the latest transformers; <code>bf16 + device_map="auto"</code> distributes Molmo2 across both 2080 Ti GPUs cleanly.	Tight version coupling to transformers/CUDA; required a careful upgrade plan to avoid breaking Ultralytics.
OpenCompass / OpenVLM Leaderboard	https://rank.opencompass.org.cn/leaderboard-multimodal	Centralized ranking of open VLMs across MMBench, MMMU, Video-MME, MVBench; used to shortlist candidate VLMs.	Benchmark scores do not always transfer to wildlife footage — we still had to verify with manual ground truth.
HuggingFace Open VLM Leaderboard	https://huggingface.co/spaces/opencompass/open_vlm_leaderboard	Same purpose as OpenCompass; cross-checked rankings.	Same benchmark-to-domain transfer caveat.

Tool	URL	Pros	Cons / Where it needed supervision
Cursor / Claude Code (Anthropic)	https://claude.com/claude-code	Used as an AI pair-programmer for prompt iteration on Molmo2, debugging the ByteTrack ID-fragmentation merger, and writing the side-by-side viewer rendering code. Hugely accelerated the Spring Quarter prototyping loop.	Needed close supervision on VLM behavior — LLM-generated code occasionally invented HuggingFace API arguments (e.g., <code>image_use_col_tokens</code>) that didn't exist in our pinned transformers version; needed to verify every API call against actual source. Also hallucinated salmon jump timestamps when reasoning about Molmo2's video output instead of running the model.
GitHub + GitHub Actions	https://github.com	Branch protection (2 reviewer approvals to merge to main), shared SSH deploy keys, thrice-daily auto-pulls on the ENGINE Lab machine.	Standard, no surprises.
Slack	https://slack.com	Async coordination with sponsor and team; easy file/screenshot sharing for visual debugging.	–

Summary of Selection Rationale

The pipeline converged on **YOLOv8 (fine-tuned) + Grounding DINO (auto-annotation) + ByteTrack (tracking) + Molmo2-8B (behavior + summary)** because that combination is the only one that simultaneously:

1. Runs on the available hardware (dual 2080 Ti for training, consumer laptop for inference).
2. Uses fully open-weights models so the open-source deliverable has no API dependency or licensing barrier.
3. Achieves >90% accuracy on the **manually-verified Katmai validation set** for bear detection.
4. Achieves the highest stage-match accuracy on our **manually-labeled feeding ground truth** for behavior classification.
5. Produces structured, parseable output that downstream feeding-event detection and environmental-data correlation can consume.

Every model choice was justified by **(a) public benchmark rankings to shortlist candidates** and **(b) Katmai-specific manual ground truth to make the final selection.**

The remaining open problem is **independent salmon detection / jump counting**, where current open VLMs are blocked by the VRAM-bound video-frame budget, and where a fine-tuned YOLO salmon detector is the most likely path forward.